Demystifying TCFD Disclosures: An AI-Powered Framework for Enhanced Transparency and Trust

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Abstract

Effective assessment of corporate climate disclosures is crucial for the global economic low-carbon transition but currently hampered by inconsistent ratings and the black-box nature of emerging AI solutions. This paper introduces a novel framework that harnesses large language models (LLMs) for transparent assessment construction rather than direct, opaque evaluation. Our methodology employs a Retrieval Augmented Generation (RAG) architecture to automatically extract granular assessment factors from corporate reports. These factors are then operationalized within an interpretable machine learning model, creating an "open box" system. Applying this framework to a global dataset of climate reports between 2020-2024, we generate macro-level insights into the evolution of disclosure practices, revealing key performance trajectories and systemic challenges. This work delivers a generalizable methodology for evaluating climate disclosure, providing actionable comparative intelligence for regulators, investors, and reporting companies.

CCS Concepts

• **Computing methodologies** → **Machine learning**; **Artificial intelligence**; *Information extraction*; • **Applied computing** → *Economics*.

Keywords

Climate finance, TCFD assessment, knowledge extraction, interpretable machine learning, banking regulation, transparency frameworks

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1 Introduction

The global economic transition towards sustainability hinges on the effective assessment of corporate climate disclosures [3]. The framework from the Task Force on Climate-Related Financial Disclosures (TCFD) [16] has become the de facto standard for such reporting. However, its widespread adoption is critically undermined by a persistent "trust deficit" in how these disclosures are evaluated. This deficit stems from three core challenges: **(1) Methodological Fragmentation**, with major consulting firms using proprietary, irreproducible assessment methods [7]; **(2) Third-Party Rating Inconsistency**, where major ESG raters show low correlation [1], eroding market confidence; and **(3) Algorithmic Opacity**, as emerging AI solutions operate as "black boxes" [10], hiding their decision-making logic. This paper confronts the standardization gap with a novel approach: we employ **AI not as an opaque assessment engine, but as an intelligent agent for transparent framework construction**. Instead of delivering a score from a black box, our methodology extracts the LLM's reasoning steps and converts them into an interpretable model. Our key contributions are as follows.

(1) **Systematic Knowledge Extraction:** We introduce a methodology using a RAG architecture to query a LLM, compelling it to deconstruct the 11 TCFD recommendations into granular, explicit assessment factors based on real-world disclosure data.
(2) **An Interpretable "Open Box" Assessment Framework:** We consolidate the extracted factors into a human-readable "knowledge table," which then serves as the foundation for training an intrinsically interpretable machine learning model. This process translates the LLM's complex reasoning into a transparent, factor-based evaluation system, moving decisively beyond the black box paradigm.
(3) **Data-Driven Insights into Global Disclosure Trends:** By applying our framework to 335 reports from 74 of the world's largest banking institutions (2020-2024), we quantify the state and trajectory of global TCFD reporting. Our analysis uncovers stark regional divergences, with European and

[1]Our code is available at https://github.com/tyyim/esg-data-analysis-dashboard. Our constructed demo can be accessed at https://esg-data-analysis-dashboard-jq3ggb763q-de.a.run.app.

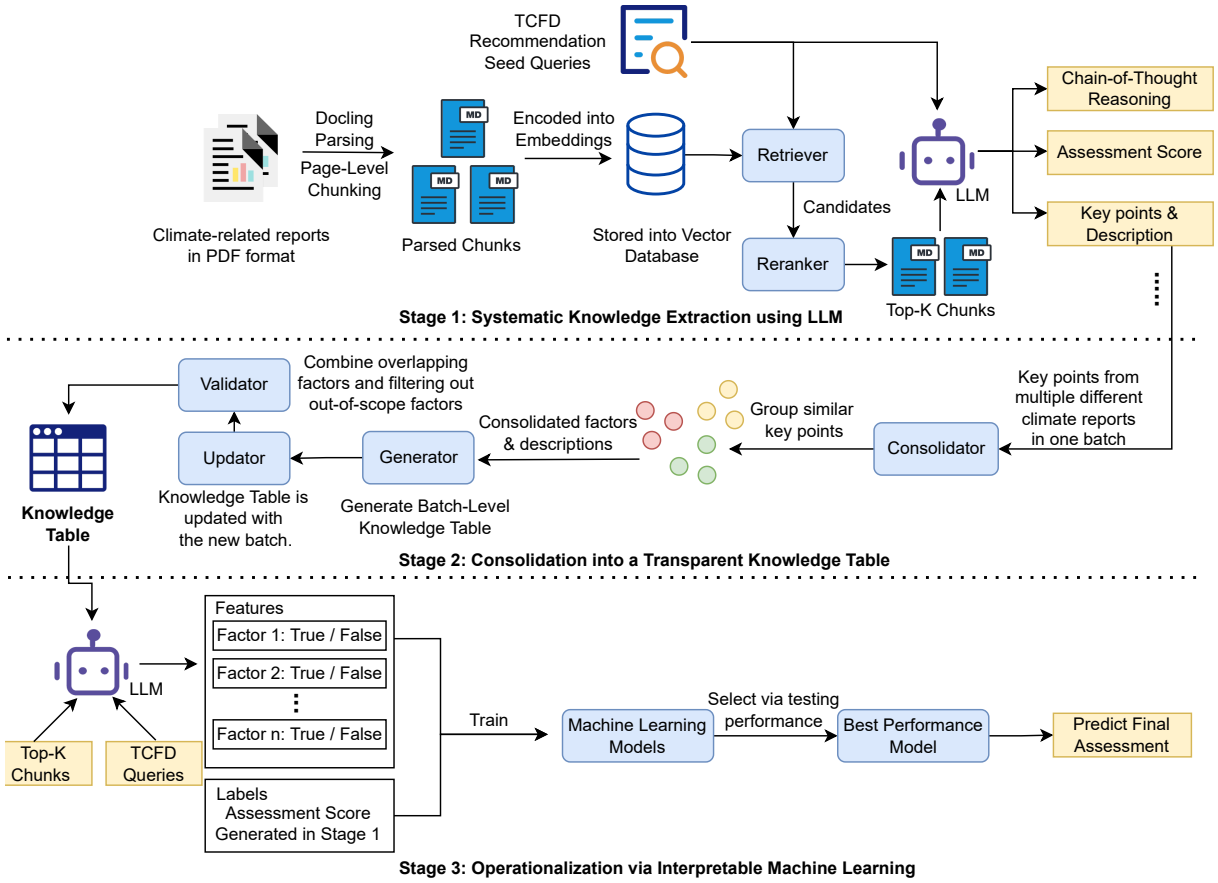


Figure 1: Overall Framework Architecture

Canadian institutions demonstrating mature, high-quality disclosures, while the US exhibits a more volatile improvement trend influenced by a shifting voluntary commitment landscape. Furthermore, our pillar-level analysis pinpoints a critical gap between procedural reporting (e.g., Governance) and substantive commitment, with "Reduction Targets" lagging significantly and representing the sector's most pressing challenge.

2 Related Work

The complexity and volume of climate-related disclosures have spurred the application of Artificial Intelligence (AI), particularly Natural Language Processing (NLP), to automate and scale analysis. The evolution of these techniques reveals a trajectory towards more sophisticated analysis, yet highlights a persistent gap in transparency and interpretability, which our work directly addresses. The analysis of climate-related disclosures has rapidly evolved from task-specific NLP models to more powerful and versatile AI architectures. While foundational efforts like the domain-specific ClimateBERT [17] highlighted the need for specialized tools, the field is now dominated by Large Language Models (LLMs) for their advanced reasoning capabilities. LLMs provide a powerful backbone for ESG analysis [19], but their inherent limitations—namely the risk of factual hallucination and reliance on static, pre-trained knowledge—make them unsuitable for direct use in high-stakes regulatory and financial contexts. To address this, Retrieval-Augmented Generation (RAG) [12, 13, 18] has emerged as the dominant paradigm. RAG enhances the reliability of LLMs by grounding their responses in externally retrieved information, ensuring that analyses are based on the specific content of a given corporate report. This approach has been successfully applied in recent ESG-focused systems like CHAT-REPORT [15], ESG-Reveal [22], and Glitter or Gold [4]. Despite these advances, a critical limitation persists: most AI-driven assessment tools function as "black boxes" [10], where the internal logic driving an evaluation remains opaque. While Chain-of-Thought (CoT) prompting [20] can make an LLM to articulate its reasoning, this explanation is not captured in a stable form. It cannot be audited or adjusted by humans. The reasoning pathways are not systematically extracted or operationalized, leaving stakeholders to trust an inscrutable process. Our research directly confronts this transparency deficit. We propose a novel methodology that is conceptually related to knowledge distillation [9], where a smaller model learns from a larger one.

3 Proposed Framework: From Knowledge Extraction to an Interpretable "Open Box" Assessment

Our framework converts opaque LLM reasoning into a transparent and structured assessment process. As shown in Figure 1, it follows a three-stage pipeline that systematically extracts, consolidates, and operationalizes the underlying logic behind disclosure evaluations.

3.1 Stage 1: Systematic Knowledge Extraction using LLM

The first stage extracts the domain knowledge embedded within an LLM to generate a set of granular assessment factors, treating the LLM as a knowledge source rather than a black-box scorer.

Contextual Grounding with RAG: To reduce hallucination and ground LLM analysis in real-world disclosures, we build a RAG system over a corpus of climate-related reports. Source documents are processed using Docling [2], which extracts both textual and tabular contents and converts them into structured, page-level Markdown chunks. Each document chunk is embedded using a pre-trained embedding model. For each query, top-n relevant chunks are retrieved based on cosine similarity and re-ranked with a reranking model. The top-k chunks are then passed to the LLM as contextual input for initial assessment generation.

Assessment Generation with Structured Reasoning: We use the 11 official TCFD recommendations as structured seed queries. For each recommendation, the LLM is provided with the query and the retrieved chunks from the corresponding climate report. The LLM is prompted to perform three tasks: (i) conduct a chain-of-thought analysis of the disclosure, assessing its comprehensiveness, quality, and gaps with respect to the TCFD recommendation; (ii) assign an alignment score ranging from 0 to 5 that reflects the overall adequacy of the disclosure; and (iii) extract structured key points grounded in the reasoning from step (i), each supported by contextual evidence from the retrieved content as key point's description. This process extracts the LLM's internal assessment rubric.

3.2 Stage 2: Consolidation into a Transparent Knowledge Table

In this stage, unstructured outputs from the initial assessment are synthesized into a structured and interpretable knowledge table.

As illustrated in Figure 1 and Algorithm 1, we adopt a sequential, batch-wise aggregation strategy to construct a generalizable knowledge table. Report-level assessments are partitioned into fixed-size batches (k reports per batch) and processed iteratively.

For each batch, we generate a knowledge table via a three-step process: (i) **key points consolidation**, which groups semantically similar keypoints and their descriptions from initial assessments; (ii) **knowledge table generation**, which summarizes each group into a distinct evaluative factor and identifies representative actions organizations have taken (or should take) to align with that factor; and (iii) **knowledge table update**, which integrates the current table with prior batches by aligning similar factors and merging their associated actions[2]. After all batches are processed, a final validation step is performed to refine the knowledge table by merging semantically redundant factors and removing out-of-scope ones. The resulting table captures the LLM's generalized assessment criteria, consolidating key evaluative dimensions and disclosure practices across diverse real-world reports.

Algorithm 1: Consolidated into a knowledge table

Input: Climate reports $\mathcal{D} = \{D_1, D_2, \ldots, D_n\}$ with keypoints $\mathcal{K}_i$ and descriptions $\mathcal{S}_i$
Output: Consolidated knowledge table $\mathcal{KT}_{\text{final}}$
Partition $\mathcal{D}$ into batches $\{\mathcal{B}_1, \ldots, \mathcal{B}_m\}$, each with k reports ;
Initialize global table: $\mathcal{KT}_{\text{global}} \leftarrow \emptyset$;
for *each batch* $\mathcal{B}_j$ **do**
  `// Step 1: Key points Consolidation`
  Group semantically similar key points into clusters $G = \{G_1, G_2, \ldots, G_n\}$;
  `// Step 2: Knowledge Table Generation`
  Initialize temporary table: $\mathcal{KT}_{\text{batch}} \leftarrow \emptyset$;
  for *each group* G_i **do**
    Summarize factor f_i from keypoints in G_i ;
    Summarize actions a_i from descriptions in G_i ;
    Add entry (f_i, a_i) to $\mathcal{KT}_{\text{batch}}$;
  `// Step 3: Knowledge Table Update`
  for *each entry* (f, a) *in* $\mathcal{KT}_{batch}$ **do**
    if *a similar factor* f^* *exists in* $\mathcal{KT}_{global}$ **then**
      Merge a into existing actions a^* ;
      Optionally refine f^* using f ;
    else
      Add (f_i, a_i) to $\mathcal{KT}_{\text{global}}$;
`// Final Knowledge Table Validation`
  Refine $\mathcal{KT}_{\text{global}}$: Merge semantically redundant factors ;
  Remove out-of-scope factors ;
return $\mathcal{KT}_{\text{final}} \leftarrow \mathcal{KT}_{\text{global}}$

3.3 Stage 3: Operationalization via Interpretable Machine Learning

The final stage operationalizes the validated knowledge table into a functional and interpretable assessment model.

Each assessment factor in the validated knowledge table is operationalized as a feature. To evaluate alignment at the feature level, we leverage the same LLM used in Stage 1 as a zero-shot text classifier, taking three inputs: (i) the original TCFD query, (ii) the retrieved content, and (iii) the knowledge table (see Figure 1). For each factor, the LLM is prompted to assess whether the retrieved disclosure content aligns with the factor's descriptions, outputting True if aligned and False otherwise. These features are then used to train interpretable machine learning models that learn to weight and integrate the assessment factors to produce an overall alignment score. Among the candidate models, we select the one with the best

[2]If processing the first batch, no update is performed; its table serves as the initial global knowledge table.

Table 1: Evaluation results of the selected best machine learning model. The table reports the test performance of the selected models across all 11 TCFD recommendations.

Model Type	TCFD Recommendations	TCFD Pillars	Accuracy	Precision	Recall	F1-score
Logistic Regression	Board's Oversight	Governance	0.818	0.863	0.818	0.830
Logistic Regression	Management Role	Governance	0.800	0.850	0.800	0.781
Logistic Regression	Climate Risks and Opportunities	Strategy	0.750	0.875	0.760	0.770
Logistic Regression	Impact of Climate Risks and Opportunities	Strategy	1.000	1.000	1.000	1.000
GBDT	Climate Scenario Analysis	Strategy	0.900	0.810	0.900	0.850
Logistic Regression	Identifying and Assessing Climate Risks	Risk Management	0.800	0.800	0.800	0.800
GBDT	Managing Climate Risks	Risk Management	0.800	0.800	0.800	0.800
Logistic Regression	Climate Risk Integration	Risk Management	0.800	0.867	0.800	0.800
GBDT	Disclosure Metrics	Metrics and Targets	0.700	0.780	0.700	0.736
Logistic Regression	Carbon Emissions	Metrics and Targets	0.780	0.930	0.780	0.810
Decision Tree	Reduction Targets	Metrics and Targets	0.700	0.933	0.700	0.761

Table 2: Report Type Distribution in Our Training Dataset

Report Type	Count
TCFD Reports	48
Climate Reports	13
Sustainability Reports	18
ESG Reports	9
Other Reports	21

performance on the testing set to generate the final assessment predictions.

This "student-teacher" approach, where an interpretable model learns the explicit logic of the "teacher" LLM, effectively translates complex reasoning into an auditable, rule-based system. The final decision is made not by the LLM, but by a transparent model that analysts can inspect and understand.

4 Experiments

To validate the proposed framework, we design experiments focused on the banking sector. The experiments include 2 parts: Training and Inference

4.1 Experimental Settings for the Training

4.1.1 Training Dataset. The dataset includes 109 publicly available reports from 24 global banking institutions between 2018 and 2024. These reports span multiple disclosure categories, including TCFD, ESG, sustainability, climate, and other integrated or non-financial disclosures. The distribution of report types is presented in Table 2.

4.1.2 Models and Parameters. For the retrieval module of our constructed RAG system, we use **gemini-embedding-001** [11] to embed both queries and document chunks into dense vector representations. The top-15 relevant chunks are retrieved and subsequently re-ranked with the **Cohere rerank-v3.5**[3]. The top-5 re-ranked chunks are then selected as contextual input for the LLM.

[3]https://cohere.com/blog/rerank-3pt5

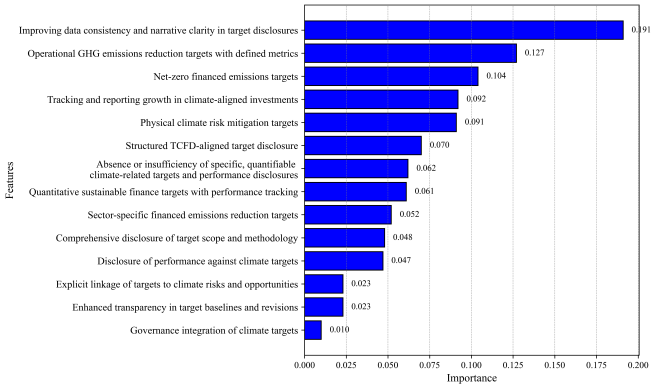


Figure 2: Feature importance of disclosure factors in the decision tree model for the *Reduction Targets* recommendation.

Gemini 2.5 Flash [6] is used for initial assessment generation (Stage 1) and factor-alignment judgment (Stage 3). **DeepSeek-R1** [8] is utilized to perform consolidation and knowledge table generation (Stage 2), where it processes unstructured outputs into a structured and validated knowledge table. Following the batch-wise aggregation strategy described in Section 3, we set the batch size to 8. All LLM calls are used a temperature of 0.0 for deterministic outputs.

4.1.3 Training. We split the dataset into a training set (90%) and a test set (10%). To address the class imbalance in the label distribution, we add class weights parameters during training. We experiment with five classifiers: Decision Tree, Logistic Regression, Support Vector Machine (SVM), Random Forest, and Gradient Boosting Decision Tree (GBDT). Model performance is evaluated using accuracy, precision, recall, and F1-score.

4.2 Training Results

Table 1 presents the testing performance of the best-performing machine learning models selected to approximate the assessment logic of our LLM agent (i.e., Gemini 2.5 Flash) across 11 TCFD recommendations. For each recommendation, we trained five classifiers

g and Assessing Climate Risks | Managing Climate Risks | Climate Risk Integration | Disclosure Metrics | Carbon Emissions | Reduction Targets | Board's Oversight

Factors Table

Factor	Aligned
Net-zero financed emissions targets	✅ Yes
Physical climate risk mitigation targets	✅ Yes
Governance integration of climate targets	✅ Yes
Structured TCFD-aligned target disclosure	✅ Yes
Disclosure of performance against climate targets	✅ Yes
Sector-specific financed emissions reduction targets	✅ Yes
Enhanced transparency in target baselines and revisions	✅ Yes
Comprehensive disclosure of target scope and methodology	✅ Yes
Tracking and reporting growth in climate-aligned investments	❌ No
Explicit linkage of targets to climate risks and opportunities	✅ Yes
Operational GHG emissions reduction targets with defined metrics	✅ Yes
Quantitative sustainable finance targets with performance tracking	❌ No
Improving data consistency and narrative clarity in target disclosures	✅ Yes
Absence or insufficiency of specific, quantifiable climate-related targets and performance disclosures	✅ Yes

Figure 3: Factors Table for Emission Targets Reduction assessment of Citigroup's 2024 Disclosures

and selected the one with the highest F1-score on the testing set. As shown in the table, the selected models achieve F1-score above 0.70 in all recommendations, with 7 out of 11 models exceeding 0.80, indicating that the models closely approximate the LLM's initial assessment.

We further analyze the internal decision logic of the trained models. Specifically, we examine the feature importance of the decision tree model trained on the "Reduction Targets" recommendation. Figure 2 shows the feature importance rankings for 14 factors derived from the knowledge table. The plot highlights the most influential disclosure factors that contribute to the decision of the LLM agent. For this recommendation, the most influential factor is "Improving data consistency and narrative clarity in target disclosures", followed by "Operational GHG emissions reduction targets with defined metrics", and "Net-zero financed emission targets".

4.3 Inference: An "Open Box" in Action

The primary contribution of this framework is its transparency, which transforms assessment from an opaque score into an actionable diagnosis. Figure 3 provides a direct visualization of this "open box" in action for the "Reduction Targets" recommendation. Instead of producing a single, unexplainable score, an analyst is presented with a clear breakdown of the underlying factors extracted by our framework. The model's evaluation is shown as a simple Yes or No against each specific factor, such as "Net-zero financed emissions targets" or "Physical climate risk mitigation targets". In this example, the framework clearly indicates that while the company has aligned on 12 out of 14 factors, it has failed to disclose adequate "Tracking and reporting growth in climate-aligned metrics" and "Quantitative sustainable finance targets with performance tracking".

This level of granular, interpretable feedback is invaluable. It allows an analyst, regulator, or the reporting company itself to understand precisely why a disclosure receives a certain evaluation and pinpoint the exact areas needing improvement. This moves beyond a simple score to provide truly actionable insights.

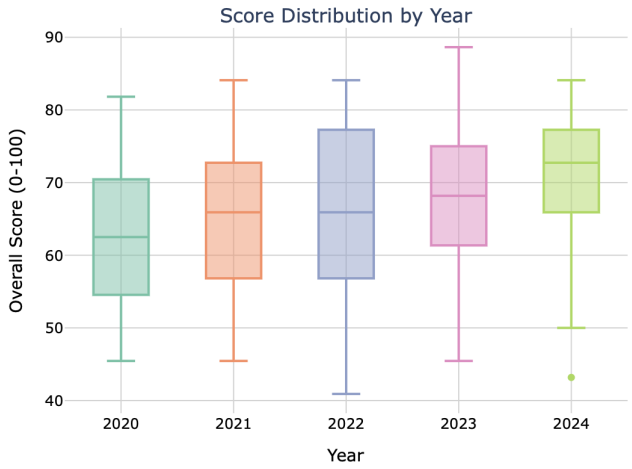


Figure 4: Company Overall Score Distribution (2020-2024)

4.4 Inference: Trend Analysis of the Assessment Results

To demonstrate the analytical power of our framework, we apply it to a large-scale inference dataset to analyze global disclosure trends. This analysis moves beyond individual reports to provide macro-level insights into the state of climate reporting. The structured, factor-based data generated by our "open box" model is aggregated to reveal performance trajectories over time, by region, and across the core TCFD pillars.

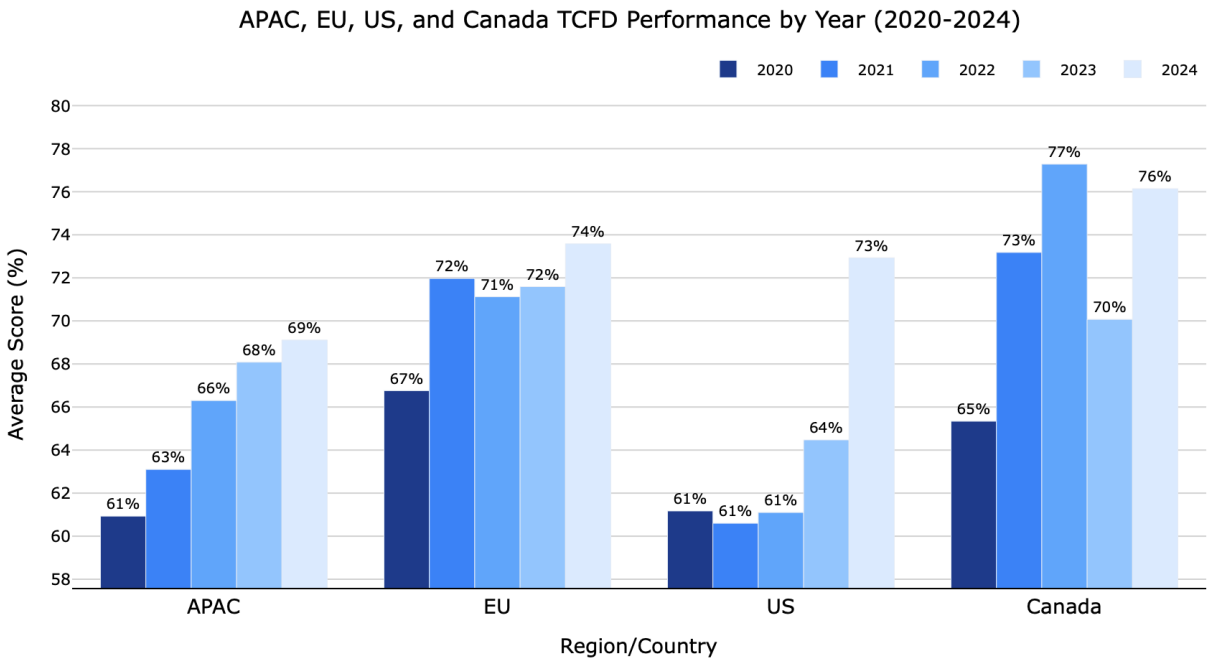


Figure 5: Regional TCFD Performance comparison of average scores from 2020 to 2024

The inference dataset comprises 335 reports (e.g., Annual, ESG, TCFD reports) from 2020 to 2024, sourced from 74 global financial institutions. Institutions are selected based on a market capitalization exceeding $50 billion USD (as of early 2025), ensuring representation from the world's leading banks. The geographical breakdown covers 17 countries, providing balanced coverage of key financial markets:

- **North America:** 27 institutions (36.5%)
- **Asia-Pacific:** 25 institutions (33.8%)
- **Europe:** 18 institutions (24.3%)
- **MENA & LATAM:** 4 institutions (5.4%)

This diverse sample, weighted towards North American and Asia-Pacific markets with significant European representation, forms a strong basis for analyzing global disclosure trends.

4.4.1 Overall and Regional Performance Trends. We aggregate the 11 recommendation scores and normalize them to produce an overall score out of 100 for each company's disclosure. The temporal analysis of disclosure scores, as illustrated in Figure 4, reveals a clear and positive trajectory from 2020 to 2024. There is a steady improvement in median disclosure quality, with the average score consistently rising over the five-year period. Reporting consistency has also improved, as shown by the narrowing inter-quartile range, with most institutions now clustering around higher scores.

However, a more notable finding is the significant divergence in performance across geographic regions, as illustrated in Figure 5. While all regions show a general trend of improvement, their trajectories differ markedly. **European Union (EU)** demonstrates consistently high performance, maintaining its position as a leader in disclosure quality with an average score of 74% in 2024. The maturity is largely underpinned by a proactive and comprehensive regulatory environment, culminating in the Corporate Sustainability Reporting Directive (CSRD), which has pushed companies towards more rigorous and standardized reporting.

The **Asia-Pacific (APAC)** region shows the most consistent and steady improvement, with its average score climbing from 61% in 2020 to 69% in 2024. Upward trend aligns with a wave of new mandatory TCFD-aligned reporting requirements being implemented in key financial hubs across the region [14].

The performance in North America presents a more complex picture and therefore we break it down into the US and Canada.

Canada stands out with leading TCFD performance, consistently outscoring the US and APAC regions and even surpassing the EU in some years. This trend indicates a strong commitment to climate disclosure among Canadian companies, reflecting a broader national focus on sustainable finance driven by both government policy and strong investor demand for transparency.

United States (US) shows a slower initial rate of improvement followed by a sharp increase in 2024. This jump may be attributable to a "survivorship bias" in the data. We observe that 8 out of 19 US companies in our dataset that had previously published dedicated sustainability or TCFD reports for 2023 ceased to do so for the 2024 reporting year. This suggests that companies continuing with detailed voluntary reporting are likely those with the strongest commitment to climate transparency, thus elevating the average score of the remaining group. This trend coincides with a shifting voluntary commitment landscape in the US, evidenced by the departure of major banking institutions from the Net-Zero Banking Alliance in early 2024, signaling a move away from globally coordinated efforts towards more individualized approaches.

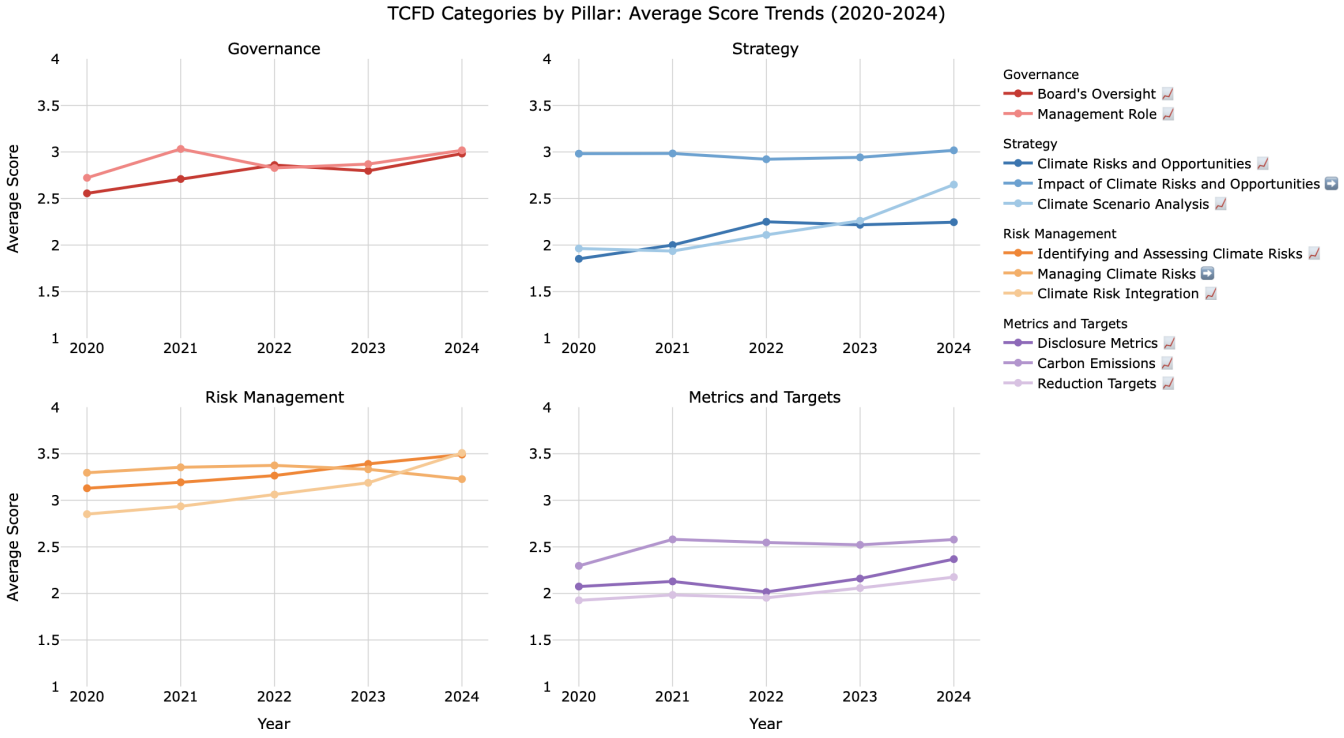


Figure 6: TCFD Categories by Pillar: Average Score Trends (2020-2024). The detailed view breaks down performance into the four core pillars, revealing specific areas of strength and weakness.

4.4.2 Pillar-Level Disclosure Analysis. An analysis of the four TCFD pillars, as shown in Figure 6, highlights both institutional strengths and persistent challenges in current disclosure practices.

The **Governance** pillar shows the most significant and consistent improvements suggesting that financial institutions have been broadly successful in establishing and reporting formal structures for board and management oversight of climate-related issues. Industry analyses show that defining roles and responsibilities is often the easiest first step for organizations to embrace climate action.

Disclosures under the **Risk Management** pillar are also a clear strength, showing high initial scores and steady improvement across all sub-categories. This likely reflects the banking sector's core competency in risk management, which institutions have been able to adapt to incorporate climate-specific risks into their existing enterprise risk frameworks.

The **Strategy** pillar shows varied performance. While disclosures on the general "Impact of Climate Risks and Opportunities" have been consistently strong, scores for identifying specific "Climate Risks and Opportunities" and assessing "Climate Scenario Analysis" started much lower. However, these weaker areas show clear and steady improvement, suggesting companies are moving from high-level statements toward more sophisticated, forward-looking strategic analysis.

Finally, the **Metrics and Targets** pillar remains the most challenging disclosure area, with scores consistently lower than those for Governance and Risk Management. While firms show improvement in reporting both "Disclosure Metrics" and "Carbon Emissions", "Reduction Targets" is a clear laggard with minimal progress. It highlights a critical gap between backward-looking measurement and forward-looking commitment, driven by the difficulty of setting credible reduction targets for complex Scope 3 financed emissions.

Collectively, these findings demonstrate how the framework can be used to move beyond simple compliance checking to generate actionable, comparative intelligence on the evolution of climate disclosure practices.

5 Practical Applications

Our framework offers practical, scalable solutions for key stakeholders, transforming climate reporting from a compliance burden into a tool for strategic action.

For Regulators and Exchanges: Despite mandates, disclosure quality is often poor, marked by "boilerplate" language and a lack of specific, quantitative metrics [21]. The issue is particularly prevalent among small and medium-sized enterprises (SMEs), which often lack the resources and expertise needed for in-depth reporting. This creates a significant monitoring burden for regulators who must sift through vast amounts of information of varying quality. Our tool automates reports screening, flags superficial disclosures and provides a systematic gap analysis for both supervisory engagement and corporate improvement. These capabilities enhance market integrity by helping to counter "greenwashing", where corporate claims may not be substantiated by concrete actions or data.

For Reporting Companies: Without the right tools, reporting becomes a 'tick-box' compliance exercise rather than a tool for strategic improvement. Our open-source framework empowers companies to perform proactive self-assessment. By identifying disclosure weaknesses against a clear methodology before publication, firms can improve their reporting quality and better integrate climate considerations into their strategy. This democratizes access to sophisticated evaluation, enabling SMEs and other resource-constrained firms to meet rising stakeholder expectations.

For Investors and Analysts: Although the volume of climate-related disclosures has increased, its decision-usefulness remains a key challenge due to inconsistency and a lack of comparability [5]. Many are forced to navigate lengthy, narrative-heavy reports to find material information and remain skeptical of unverified corporate claims. Our framework acts as an analytical intermediary, translating unstructured reports into transparent and comparable insights. It enables the rapid assessment of disclosure quality across a portfolio, helping investors allocate capital more effectively and focus shareholder engagement on specific, material weaknesses.

6 Limitations

While our "open-box" approach represents a meaningful step toward transparency, it has several limitations. First, the framework's performance is fundamentally dependent on the upstream Retrieval-Augmented Generation (RAG) system. The quality of the final assessment is contingent on the RAG module's ability to retrieve the most relevant text chunks, and failures in this retrieval process can lead to a flawed analysis.

Second, the framework is designed to distill the logic of a "teacher" LLM, not to establish an objective ground truth. This means our model learns to mimic the reasoning of a proprietary, non-auditable system, faithfully replicating any of its inherent biases, inconsistencies, or logical gaps.

Finally, a tension exists between interpretability and adjustability. Although the framework produces a transparent model, its logic is statistically optimized during training. Manually adjusting the learned rules would break the model's statistical integrity, requiring a complete retraining to formally incorporate expert-driven changes.

7 Future Work

Future work will focus on enhancing the framework's adaptability, interactivity, and analytical depth across three key areas:

First, we will adapt the framework to align with emerging global standards, primarily the IFRS S2 Climate-related Disclosures. This requires sourcing a new corpus of IFRS S2-aligned reports to serve as a reference dataset, which is essential for grounding the RAG system and enabling the LLM to extract a relevant assessment framework for the new standard.

Second, we plan to develop a more powerful human-in-the-loop mechanism to better integrate expert judgment. It will involve a two-stage process where experts first curate the LLM-generated knowledge table to guide model training, and then validate the resulting logic of the final interpretable model, creating a hybrid assessment framework.

Finally, to improve analytical reliability, we will move beyond the current RAG system to an advanced agentic reasoning approach. The future system will be designed to self-critique its findings and autonomously perform iterative retrieval cycles if information is insufficient, allowing it to either find missing details with greater certainty or confirm that a disclosure is genuinely deficient.

8 Conclusions

The paper confronts the critical "trust deficit" in climate disclosure by fundamentally re-imagining the role of AI in financial oversight. Instead of using Large Language Models as opaque, black-box judges, we leverage them as intelligent partners in constructing a transparent assessment framework. Our key innovation is a hybrid architecture that distills the nuanced, contextual reasoning of a state-of-the-art LLM into an intrinsically interpretable machine learning model. The result is a fully explainable "open box" where the logic of evaluation is clear, auditable, and traceable from high-level recommendation to granular factor.

Our application of this framework provides not just a validation of the tool, but also crucial intelligence for the financial community. The findings—a global improvement in reporting quality overshadowed by regional disparities and a systemic weakness in forward-looking "Metrics and Targets"—offer a data-driven diagnosis of the current state of climate disclosure. This evidence confirms that while foundational governance and risk management structures are being established, the critical step of setting credible, quantitative reduction targets remains a major hurdle for the industry.

The implications of this methodology are immediate and far-reaching. It provides regulators, investors, and reporting companies with a scalable and auditable tool to move beyond simple compliance and drive meaningful improvement. As the financial world adopts more stringent standards like IFRS S2, such transparent systems will be essential for maintaining market integrity. By making the *why* behind an assessment as important as the score itself, our research demonstrates a viable path toward a new era of oversight—one where AI fosters trust not through inscrutable authority, but through unambiguous clarity, helping to build a more accountable and sustainable global economy.

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